MAINCRAFTS TECHNOLOGY

Cybersecurity & Ethical Hacking Internship

TASK - 2

Build Your Personal Cybersecurity Lab

Lab Build Report with Evidence

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1. Objective

The objective of this task was to build a safe and isolated cybersecurity practice environment on a personal computer. The lab serves as the foundation for later work such as vulnerability assessment, web application testing, exploitation practice and incident response. It consists of a Kali Linux attacker virtual machine that also hosts an intentionally vulnerable target (OWASP Juice Shop, running in Docker), connected through a segmented NAT plus Host-Only network that keeps lab traffic away from real systems.

2. Lab Architecture

The lab runs entirely inside Oracle VirtualBox on the host computer. A single Kali Linux VM acts as the attacker machine and also hosts the target: OWASP Juice Shop runs as a Docker container published on port 3000. The VM has two network adapters. The NAT adapter (eth0) is used for updates and downloads, and the Host-Only adapter (eth1, 192.168.56.10/24) carries the lab traffic.

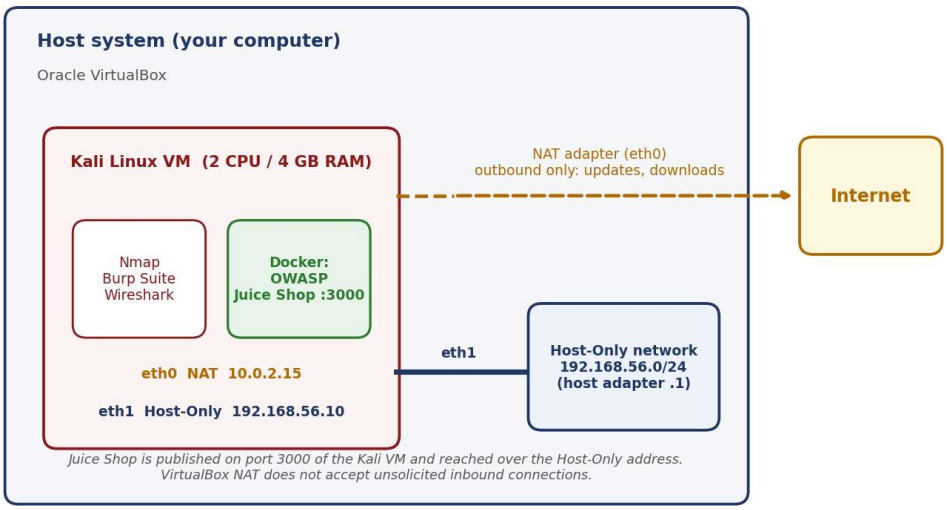


Figure 1: Lab network architecture (NAT for updates, Host-Only for lab traffic)

2.1 Network Design

Network	Purpose	Attached To
NAT	Outbound internet access for OS updates and for downloading tools and the Docker image	Kali Linux Adapter 1 (eth0, 10.0.2.15/24)
Host-Only (192.168.56.0/24)	Isolated lab traffic; Juice Shop is reached on 192.168.56.10:3000	Kali Linux Adapter 2 (eth1, 192.168.56.10/24)

3. Implementation

Step 1: Install VirtualBox and Create the Host-Only Network

VirtualBox was downloaded from the official site and installed. A Host-Only network was then created in the VirtualBox network manager. The network used by the lab is VirtualBox Host-Only Ethernet Adapter #2, with the host adapter at 192.168.56.1/24. Its DHCP server is disabled, so the Kali VM was given a static address in Step 4.

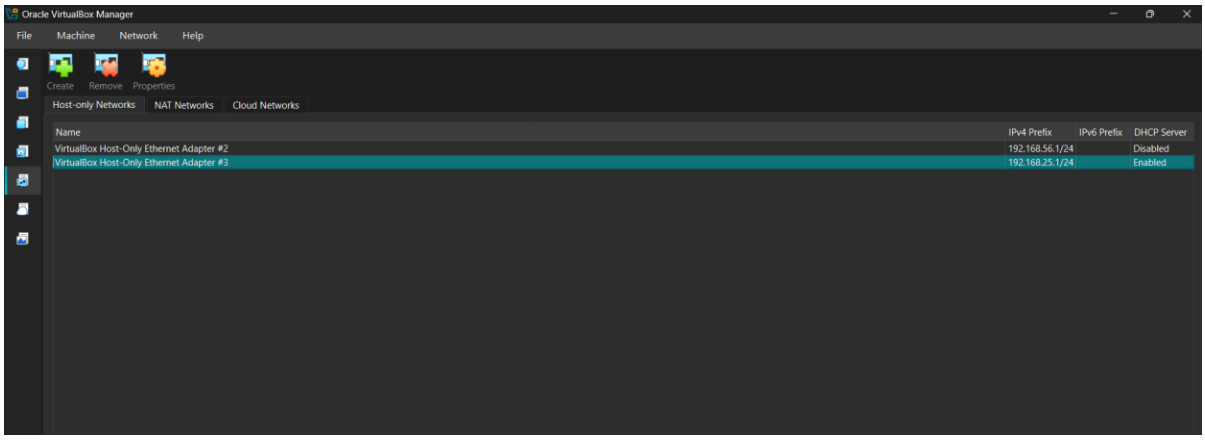


Figure 2: VirtualBox Host-only Networks: Adapter #2 (192.168.56.1/24, DHCP disabled) is used by the lab

Step 2: Deploy Kali Linux (Attacker VM)

Kali Linux (kali-rolling) was deployed as a VirtualBox VM and configured as follows:

- 2 CPUs and 4 GB (4096 MB) RAM
- Adapter 1: NAT (internet access for updates)
- Adapter 2: Host-Only, attached to VirtualBox Host-Only Ethernet Adapter #2 (lab traffic)

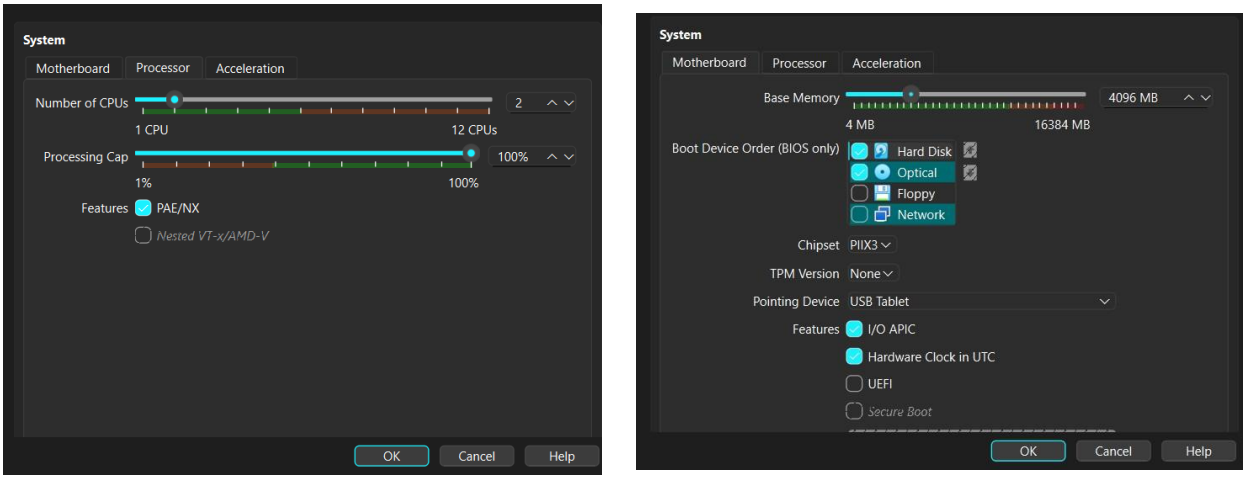


Figure 3: Kali VM settings: 2 CPUs (left) and 4096 MB base memory (right)

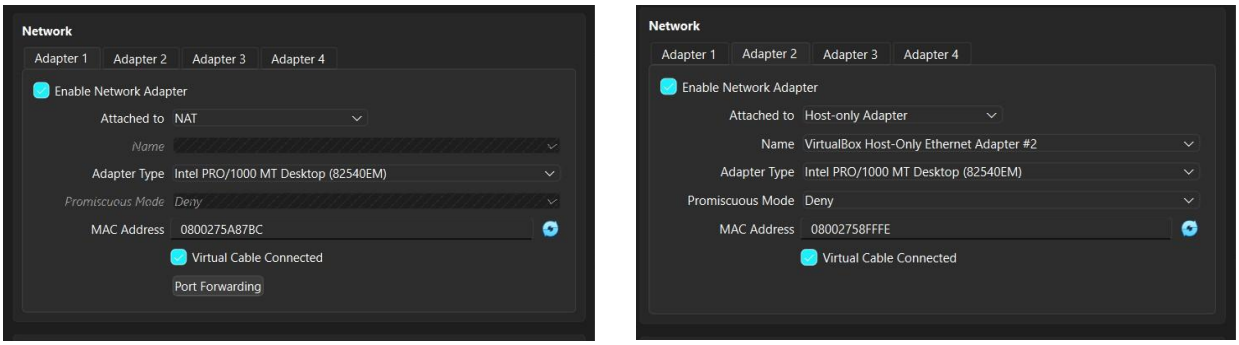


Figure 4: Kali VM network adapters: Adapter 1 = NAT (left), Adapter 2 = Host-only (right)

After booting, the system was updated and a working folder structure was created for evidence:

```
sudo apt update && sudo apt upgrade -y
mkdir -p ~/lab/{scans,notes,pcaps,screenshots}
ls ~/lab
```

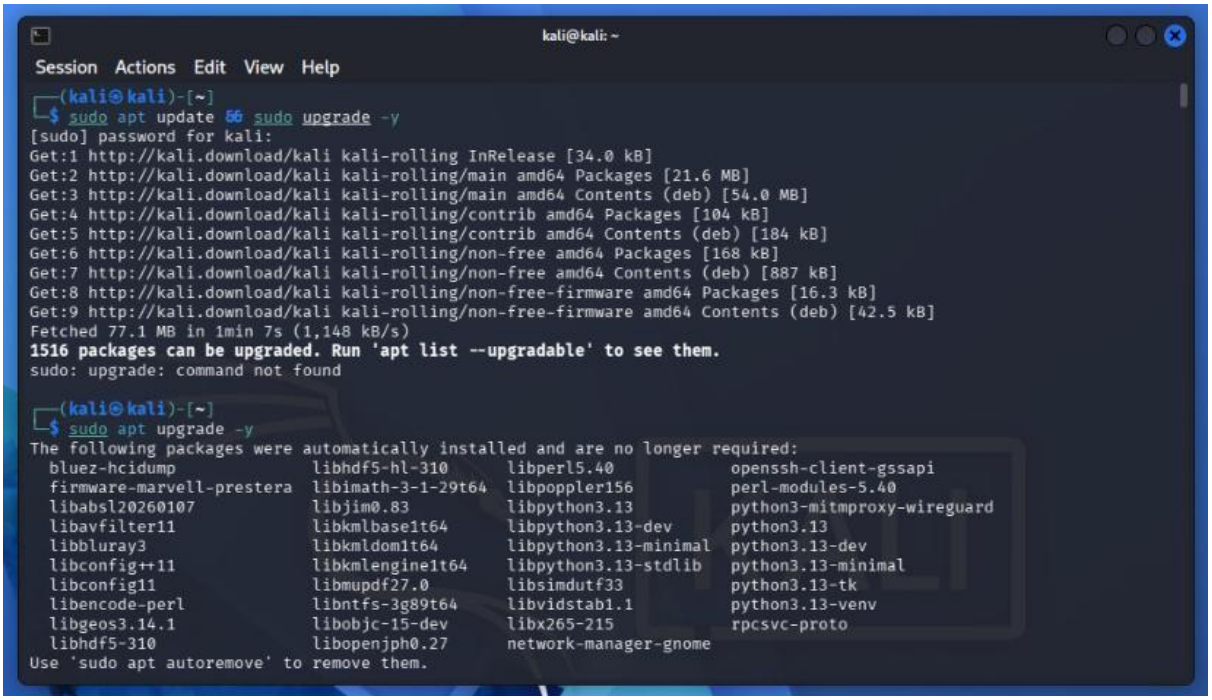


Figure 5: System update on Kali (the first attempt had a typo and was re-run correctly; see Section 6)

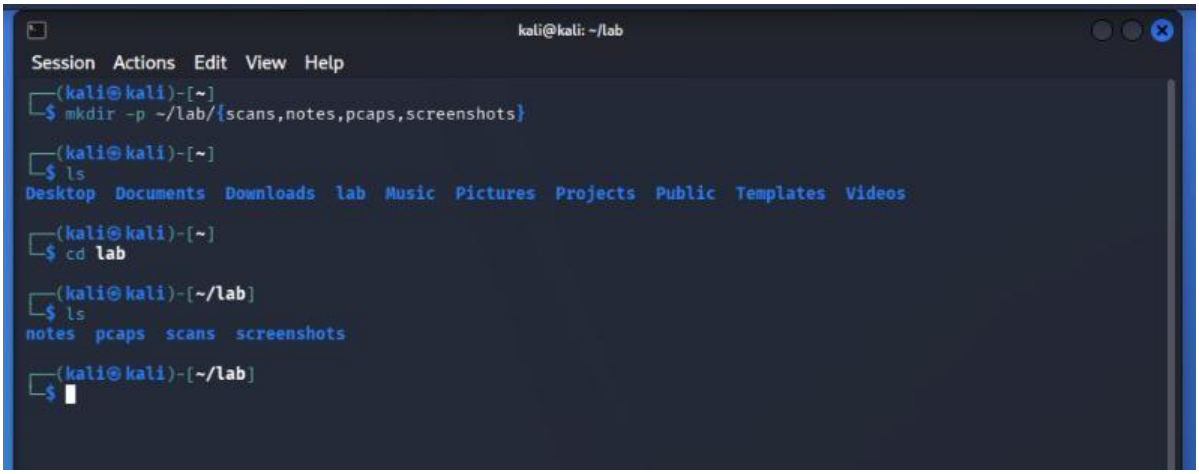


Figure 6: Lab working folders created under ~/lab (notes, pcaps, scans, screenshots)

Step 3: Deploy the Vulnerable Target

Target chosen: OWASP Juice Shop. It runs as a Docker container on the Kali VM itself, published on port 3000, and is reached through the Kali VM's Host-Only address (192.168.56.10) or at http://localhost:3000.

Juice Shop via Docker

```
sudo apt install -y docker.io
sudo systemctl enable --now docker
sudo docker pull bkimminich/juice-shop
sudo docker run -d -p 3000:3000 --name juice bkimminich/juice-shop
sudo docker ps
```

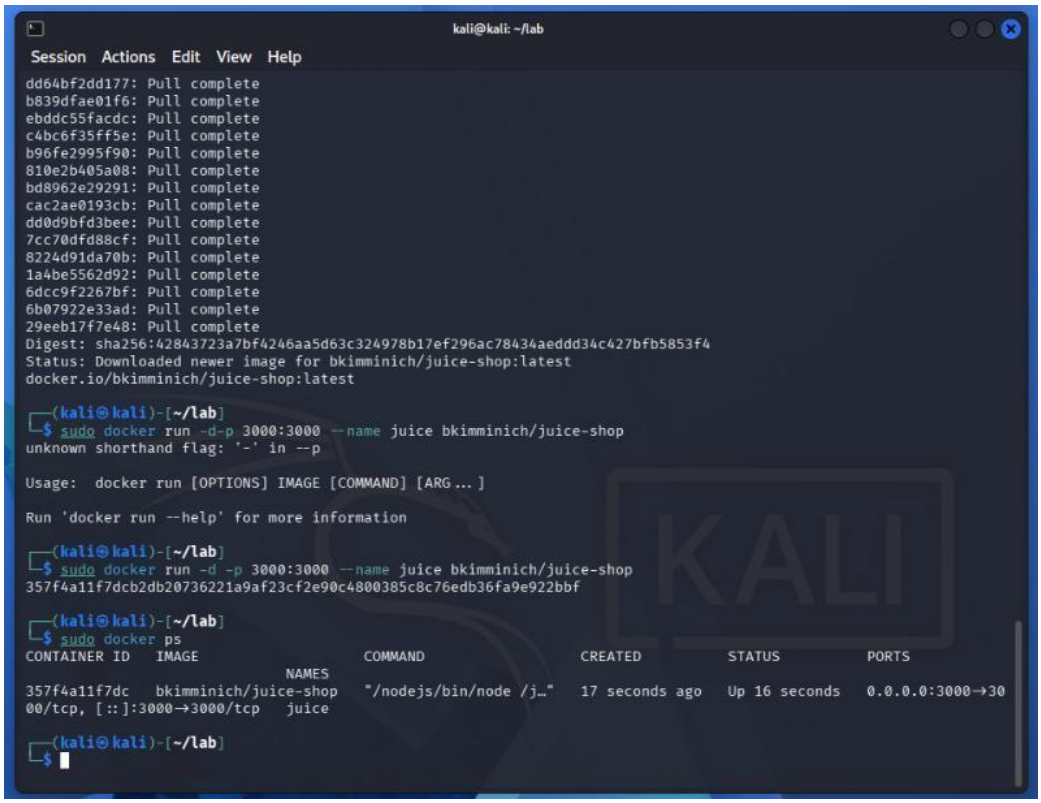


Figure 7: Juice Shop image pulled and the juice container running (docker ps shows Up, port 3000). The first docker run had a typo and was re-run; see Section 6

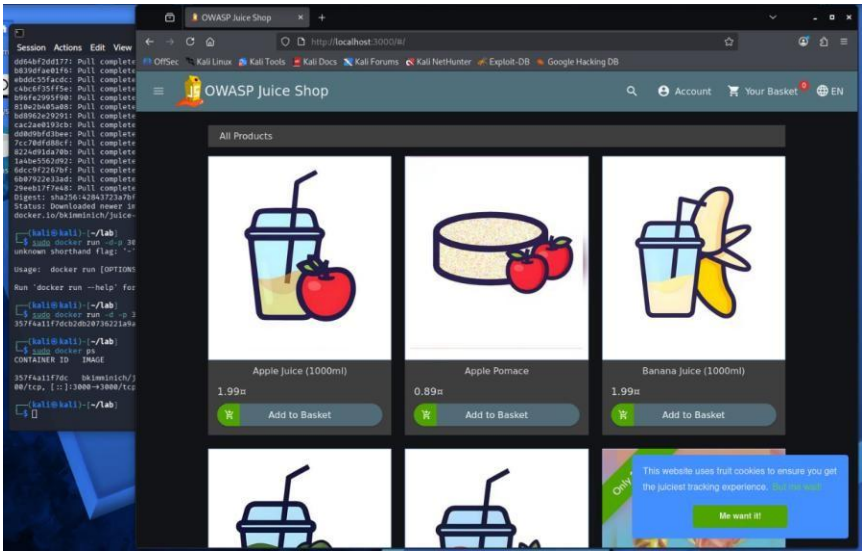


Figure 8: OWASP Juice Shop home page opened in Firefox at http://localhost:3000

Step 4: Verify Connectivity

The IP configuration of the attacker was checked, the lab address was pinged, and a ping sweep of the Host-Only subnet was run. The Host-Only interface (eth1) had no address because the network's DHCP server is disabled, so it was given the static address 192.168.56.10/24.

```
ip a
ping -c 4 192.168.56.10
nmap -sn 192.168.56.0/24
```

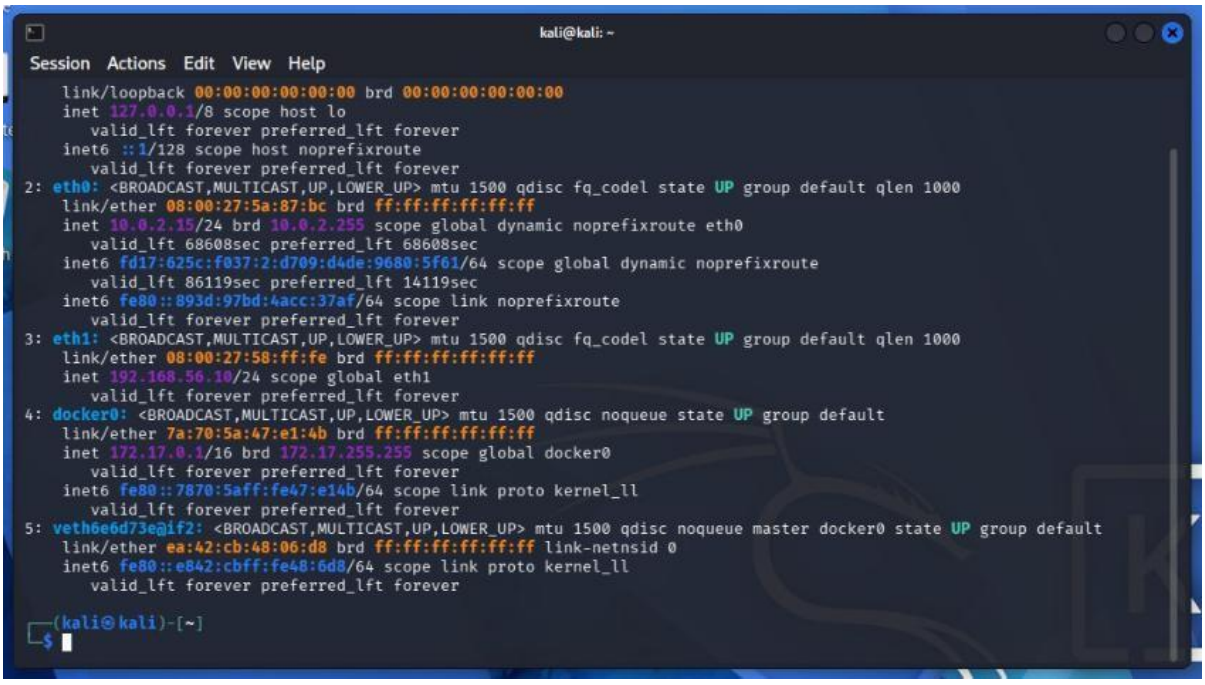


Figure 9: ip a output: eth0 (NAT, 10.0.2.15/24) and eth1 (Host-Only, 192.168.56.10/24)

The ping returned 4 of 4 replies with 0% packet loss. The sweep of 192.168.56.0/24 found 2 hosts up: the VirtualBox host adapter (192.168.56.1) and the Kali VM (192.168.56.10).

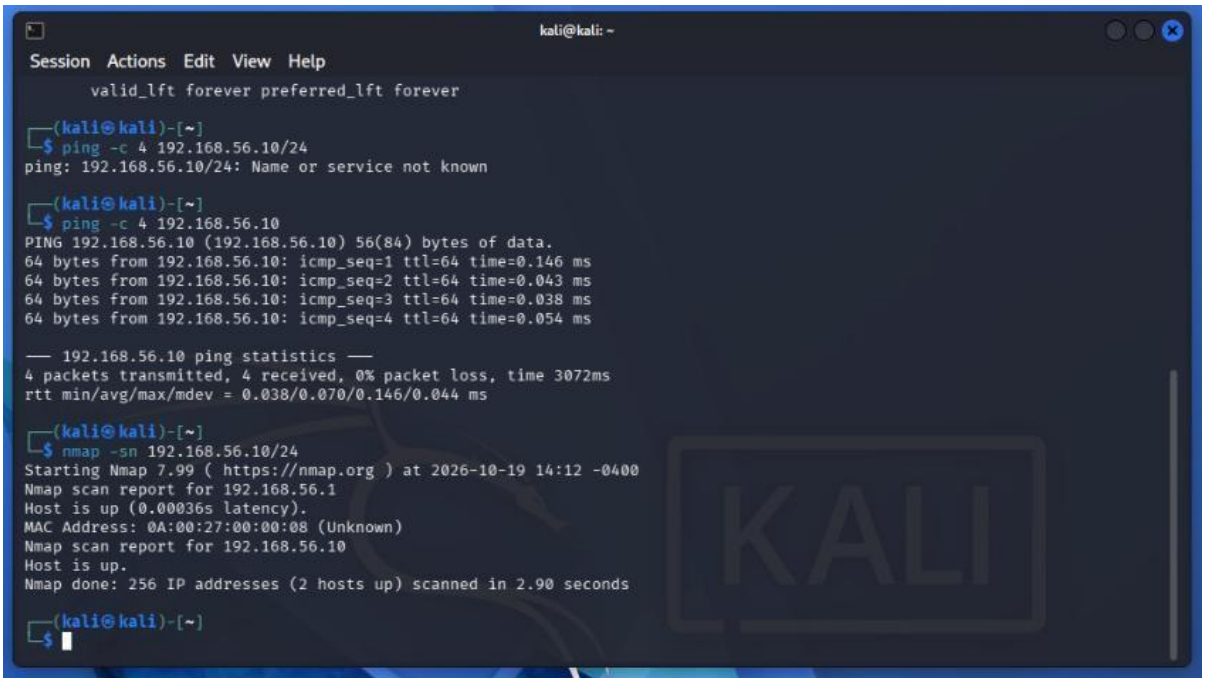


Figure 10: Ping replies from 192.168.56.10 (0% loss) and nmap -sn host discovery (2 hosts up). The first ping failed because of a typo; see Section 6

Step 5: Basic Validation

Two checks were performed to confirm the lab is functional.

- Burp Suite: the browser was routed through Burp's proxy and HTTP requests to the lab host (192.168.56.10) were intercepted with Intercept on.

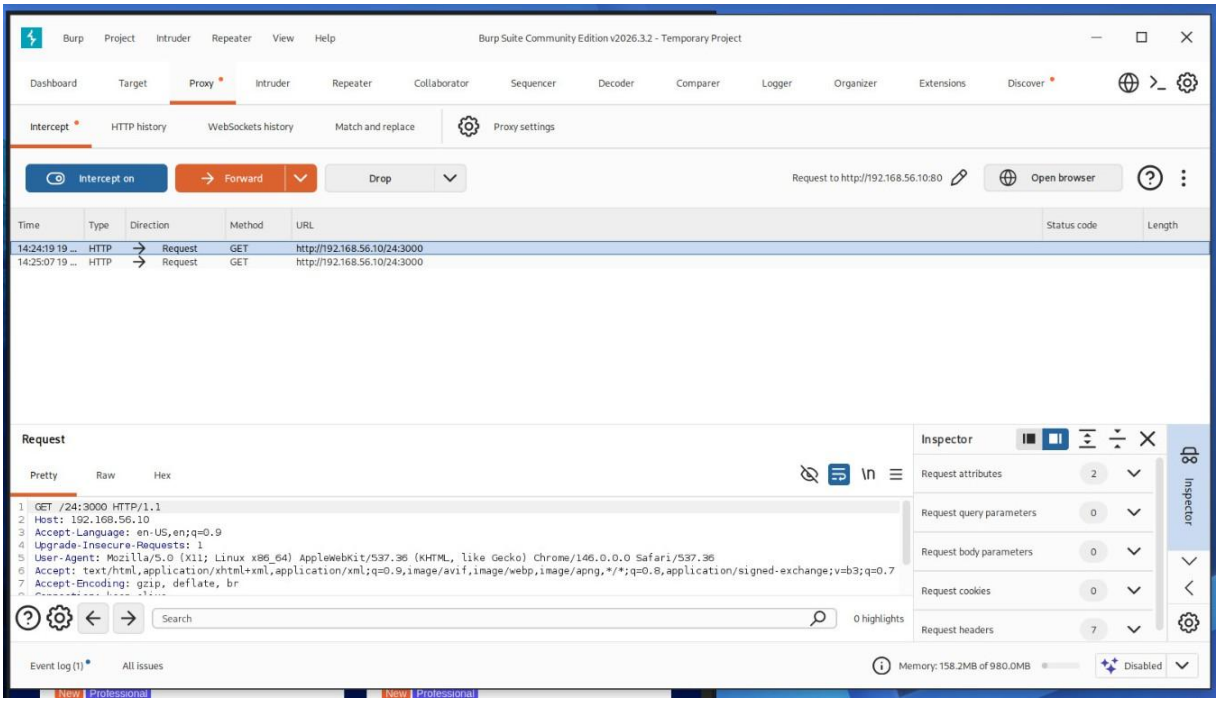


Figure 11: Burp Suite Proxy > Intercept (Intercept on) capturing HTTP requests to the lab host 192.168.56.10

- Wireshark: traffic between the browser and Juice Shop was captured on the loopback interface (lo), because the target runs on the same VM as Kali. Follow TCP Stream shows the plain-text HTTP request and response.

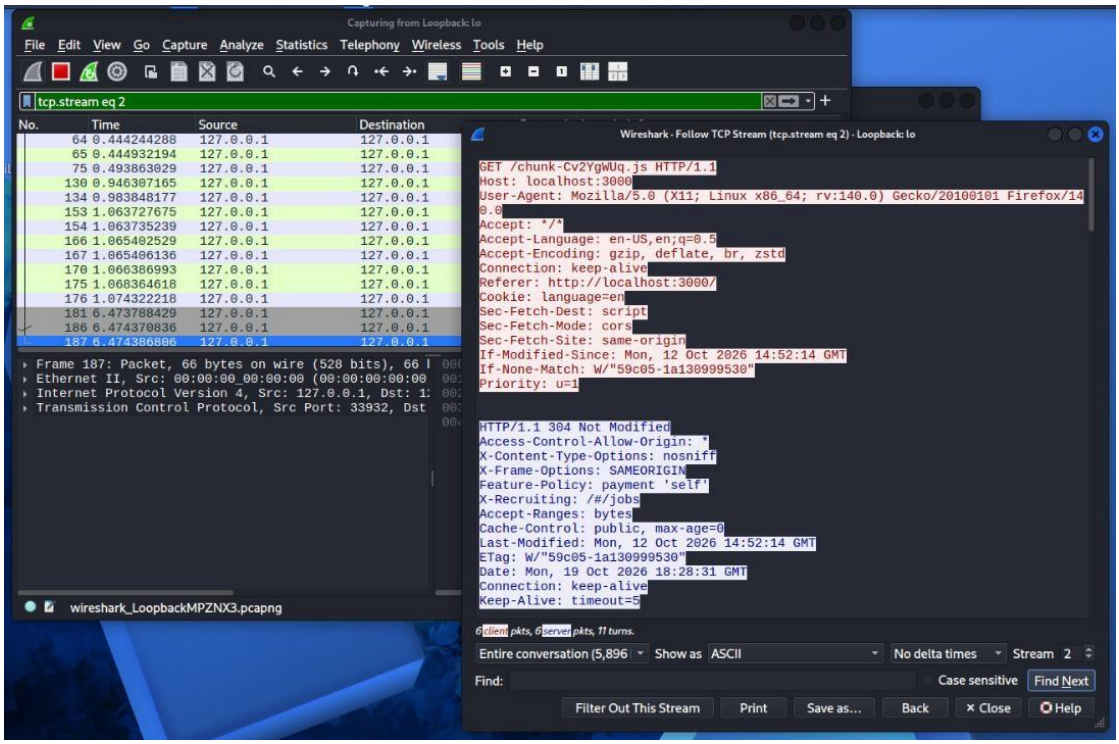


Figure 12: Wireshark capture on the loopback interface (lo) with Follow TCP Stream showing the HTTP request/response to Juice Shop (localhost:3000)

4. Network Configuration Summary

Item	Value
Kali Adapter 1	NAT (eth0), IP: 10.0.2.15/24
Kali Adapter 2	Host-Only (eth1), IP: 192.168.56.10/24 (static)
Target IP (Host-Only)	192.168.56.10 (Juice Shop container on the Kali VM)
Host-Only subnet	192.168.56.0/24 (host adapter 192.168.56.1, DHCP disabled)
Target service and port	OWASP Juice Shop (Docker) on TCP port 3000

5. Lab Isolation and Safety

Because this lab deliberately runs vulnerable software and offensive tools, isolation was treated as a core requirement:

- The Kali VM has only two network adapters: NAT and Host-Only. There is no bridged adapter, so the vulnerable Juice Shop container is not exposed to the home or college network.
- The NAT adapter is used only for updates and downloads (apt packages and the Docker image). VirtualBox NAT does not accept unsolicited inbound connections by default.

- Juice Shop runs inside a Docker container, which adds a further layer of separation from the Kali operating system.
- All scanning was limited to the lab range 192.168.56.0/24 and the local host. No external systems were tested.
- VM snapshots can be taken before testing so the lab can be restored to a clean state after experiments.

6. Challenges and Troubleshooting

The issues below were met while building the lab and how each was resolved:

Issue	Cause	Resolution
sudo: upgrade: command not found	The second apt was left out of the chained update command	Re-ran sudo apt upgrade -y (Figure 5)
unknown shorthand flag: '-' in -p	Missing space between -d and -p in the docker run command	Re-ran with -d -p 3000:3000; the container started (Figure 7)
eth1 had no IP address	DHCP server is disabled on Host-Only Adapter #2 (Figure 2)	Assigned the static address 192.168.56.10/24 to eth1
ping: 192.168.56.10/24: Name or service not known	The /24 suffix was included in the ping target; ping needs a plain IP address	Pinged 192.168.56.10: 4/4 replies, 0% loss (Figure 11)

7. Conclusion

A working, isolated cybersecurity lab was built, consisting of a Kali Linux VM that hosts OWASP Juice Shop in Docker and a segmented NAT plus Host-Only network. Connectivity was verified with ping and Nmap, and the toolchain was validated with Nmap, Burp Suite and Wireshark. This lab provides a safe base for the upcoming tasks on vulnerability assessment, web application testing and incident response.